

(0000CJA100120029)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)

ALLEN COMMON TEST

01-11-2020

JEE(Main+ Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV, V, VI, VII, VIII & IX]**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	D	C	A	B	B	A,C,D	A,B,C	B,C,D	A,C,D
SECTION-II	Q.	11	12	3	4	5	6	7	8	9	10
	A.	B,D	B,D	3	4	5	6	7	8	9	10
	A.	1.58	1.60	5.00	6.72	0.75	9.80	7	8	9	10

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	D	D	A	A	A,B,C,D	A,B,C	A,B,C,D	A,B,C
SECTION-II	Q.	11	12	3	4	5	6	7	8	9	10
	A.	B,D	A,B,C	3	4	5	6	7	8	9	10
	A.	2.00	5.00	0.25	240.00	9.00	7.00	7	8	9	10

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	D	B	B	A	D	B,C,D	A,B,C,D	B,D	A,B,D
SECTION-II	Q.	11	12	3	4	5	6	7	8	9	10
	A.	A,B,C	B,C	3	4	5	6	7	8	9	10
	A.	2.00	51.50	4.50	0.90	1.50	1.50	7	8	9	10

(0000CJA100120030)

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JEE(Main+ Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV, V, VI, VII, VIII & IX]**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C	B,C	A,D	A,B,D	A,B,C	A,B,C	7	8	9	10
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	0.25	2.12	2.82	12.00	7.66	0.02	7	8	9	10
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	3	4	5	3	2	1	7	8	9	10

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C	A,D	A,B,C	A,B,D	A,C	A,C,D	7	8	9	10
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	50.00	7.50	6.00	3.00	5.00	6.00	7	8	9	10
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	3	5	2	9	2	6	7	8	9	10

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B	A,C	A,C	B,D	A,C	C	7	8	9	10
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	0.33	8.00	2.18 or 2.19	2.00	19.90	2.00	7	8	9	10
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	1	3	1	2	4	3	7	8	9	10



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PAPER-1

PART-1 : PHYSICS

SOLUTION

SECTION-I

1. **Ans. (D)**

Sol. Let friction on tyre is f ,

for up hill motion

$$f - mg \sin \theta - kv_1^2 = 0 \quad \dots(i)$$

for down hill motion

$$f + mg \sin \theta - kv_2^2 = 0 \quad \dots(ii)$$

from (i) & (ii)

$$2f = k(v_1^2 + v_2^2)$$

for flat road

$$f - kv^2 = 0$$

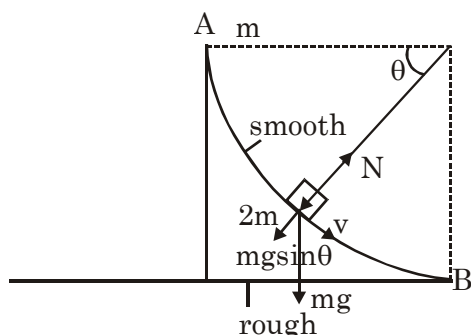
$$v^2 = \frac{v_1^2 + v_2^2}{2} = \frac{12^2 + (16)^2}{2}$$

$$v = \sqrt{200} = 10\sqrt{2} = 14.14$$

$$\approx 14 \text{ km/hr}$$

2. **Ans. (D)**

Sol.



By w.e.t.

$$mg \ell \sin \theta = \frac{1}{2}mv^2$$

$$v^2 = 2g \ell \sin \theta$$

$$N - mg \sin \theta = \frac{mv^2}{\ell}$$

$$N = 3mg \sin \theta$$

Horizontal force by block on the wedge

$$F = N \cos \theta = 3mg \sin \theta \cos \theta$$

$$F = \frac{3}{2}mg \sin 2\theta$$

$$F_{\max} = \frac{3}{2}mg = \text{friction force.}$$

3. **Ans. (C)**

$$\text{Sol. } v = \frac{30}{1 + \frac{t}{10}} \Rightarrow 1 + \frac{t}{10} = \frac{30}{v}$$

$$\frac{1}{10} = -\frac{30}{v^2} \frac{dv}{dt}$$

$$\frac{dv}{dt} = -\frac{v^2}{300} \text{ - Non uniform acceleration}$$

$$\text{when } v = 10 \quad a = -\frac{1}{3} \text{ m/s}^2$$

$$\text{when } v = 15 \quad a = -\frac{15 \times 15}{300} = -\frac{15}{20}$$

$$F = ma = 60 \times \frac{15}{20} = 45 \text{ N}$$

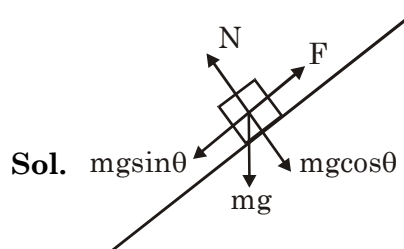
$$\text{when } v = 10 \quad a = \frac{1}{3}$$

$$F = ma$$

$$F = 60 \times \frac{1}{3} = 20 \text{ N}$$

$$P = Fv = 200 \text{ watt.}$$

4. **Ans. (A)**



Sol. $mgsin \theta$

maximum value of friction

$$f_{\max} = \mu mg \cos \theta = 2 \tan \theta mg \cos \theta = 2mg \sin \theta$$

Hence maximum value of F up the incline, force for zero acceleration

$$F = mgsin \theta + f_{\max}$$

$$F = 3mg \sin \theta$$

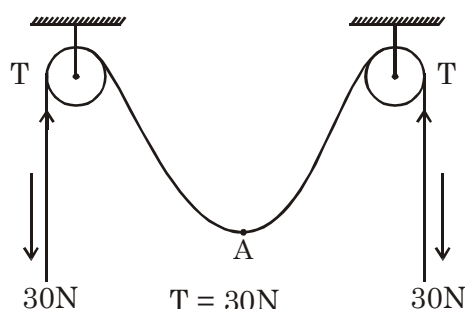
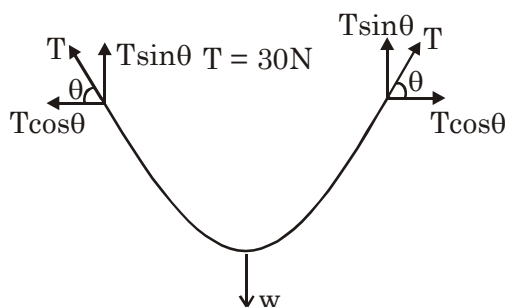
Down the incline force, for zero acceleration

$$F = f_{\max} - mgsin \theta$$

$$= mg \sin \theta$$

5. Ans. (B)

Sol.



$$2T \sin \theta = 40$$

$$\sin \theta = \frac{40}{60} = \frac{2}{3}$$

Tension at mid point $T \cos \theta$

$$= 30 \frac{\sqrt{5}}{3} = 10\sqrt{5} \text{ N}$$

6. Ans. (B)

Sol. for first one second $v = 10 \times 1 = 10 \text{ m/s}$

$$h = \frac{1}{2}gt^2 = 5 \text{ m}$$

distance travelled in next 5sec = $10 \times 5 = 50 \text{ m}$

total time = 8 sec

time left = 2 sec

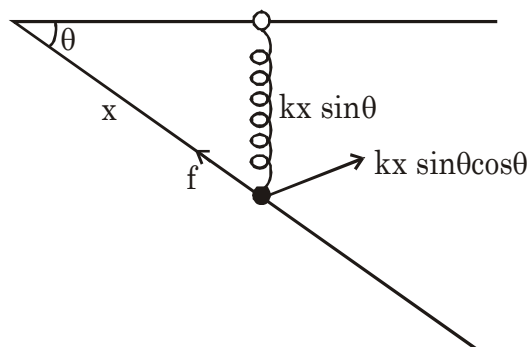
$$\text{distance in next 2 sec} = 10 \times 2 + \frac{1}{2}(0 \times 2)$$

$$= 40 \text{ m}$$

Total height = 95 m

7. Ans. (A,C,D)

8. Ans. (A,B,C)



Sol.

$$f = \mu N$$

$$f = \mu kx \sin \theta \cos \theta$$

By W.E.T.

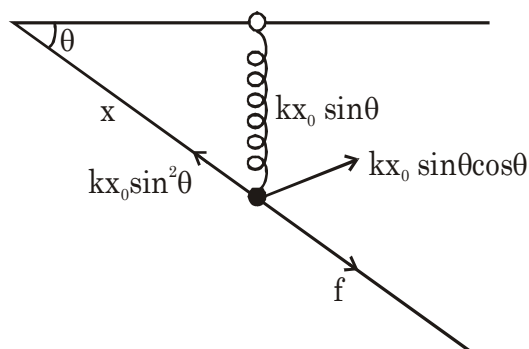
$$W_f + W_{sp} = \Delta KE$$

$$-\int_0^{x_0} \mu kx \sin \theta \cos \theta dx - \int_0^{x_0} kx \sin^2 \theta dx$$

$$= 0 - \frac{1}{2}mv_0^2$$

$$x_0 = \sqrt{\frac{mv_0^2}{k(\sin^2 \theta + \mu \sin \theta \cos \theta)}}$$

when bead comes to rest



block will come to vertex again if

$$kx_0 \sin^2 \theta > \text{friction}$$

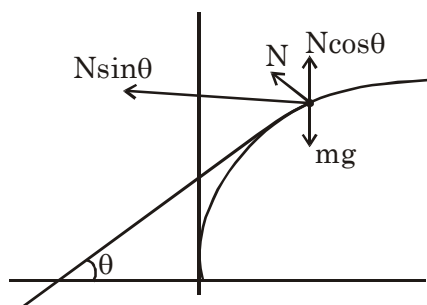
$$kx_0 \sin^2 \theta > \mu kx_0 \sin \theta \cos \theta$$

$$\tan \theta > \mu$$

9. Ans. (B,C,D)

Sol. $mg - N \cos \theta = ma_y$

$$N \cos \theta = m(g - a_y)$$



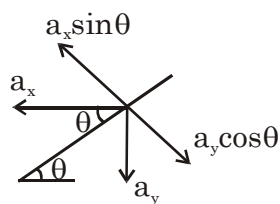
$$N \sin \theta = m a_x$$

$$\tan \theta = \frac{a_x}{g - a_y}$$

$$\frac{a_x}{10 - 6} = 1 \Rightarrow a_x = 4 \text{ m/s}^2$$

$$a_c = a_y \cos \theta - a_x \sin \theta$$

$$= (6 - 4) \frac{1}{\sqrt{2}} = \sqrt{2}$$



$$a_c = \frac{v^2}{R} \Rightarrow R = \frac{v^2}{a_c} = \frac{4}{\sqrt{2}} = 2\sqrt{2} \text{ m}$$

$$N \sin \theta = m a_x$$

$$N \times \frac{1}{\sqrt{2}} = 2 \times 4$$

$$N = 8\sqrt{2} \text{ Newton.}$$

10. **Ans. (A,C,D)**

Sol. for $\theta \leq 45^\circ$ friction is static hence contact force $R = mg = 20 \text{ N}$

$\theta = 45^\circ$ - angle of repose

$$\mu = \tan \theta = 1$$

$$\text{at } 53^\circ a = g (\sin \theta - \mu \cos \theta) = 2 \text{ m/s}^2$$

11. **Ans. (B,D)**

Sol. A moving along a path in which time is minimum

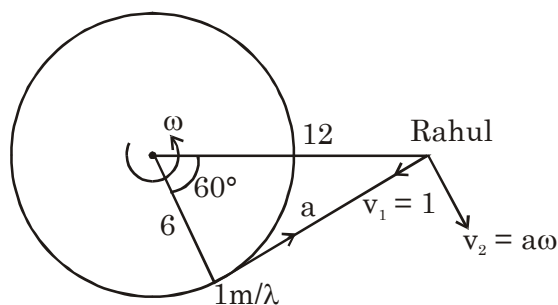
B - moving along zero drift path.

12. **Ans. (B,D)**

Sol. Angular velocity of Priya $\omega = \frac{v}{r}$

$$\omega = \frac{1}{6} \text{ rad/sec.}$$

as Priya moving on circle w.r.t. Rahul with angular velocity ω . Angular velocity of Rahul is also equal and in opposite sense w.r.t. Priya. So Rahul has two components of velocity w.r.t. Priya one along line of joining another perpendicular to line of joining.



$$v_2 = 12 \sin 60^\circ \omega = \sqrt{3}$$

Velocity of Rahul w.r.t. Priya

$$= \sqrt{1^2 + (\sqrt{3})^2} = 2 \text{ m/s}$$

SECTION-II

1. **Ans. 1.58**

Sol. Maximum speed is at equilibrium let displacement of block is x in equilibrium $4kx = mg$

$$x = \frac{mg}{4k}$$

Apply W.E.T.

$$mgx + \frac{1}{2}k(0 - (2x)^2) = \frac{1}{2}mv^2$$

$$\frac{m^2g^2}{4k} - \frac{k}{2} \frac{4m^2g^2}{16k^2} = \frac{1}{2}mv^2$$

$$\frac{m^2g^2}{8k} = \frac{1}{2}mv^2$$

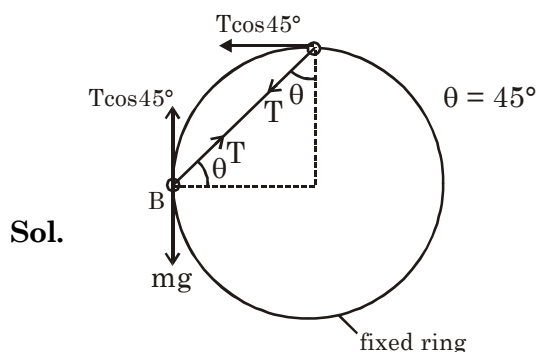
$$v = g\sqrt{\frac{m}{4k}} = 10\sqrt{\frac{2}{4 \times 20}}$$

$$v = \frac{3.16}{2} = 1.58 \text{ m/s}$$

2. **Ans. 1.60**

$$\text{Sol. } \frac{t_1 + t_2}{5} = \frac{3 + 5}{5} = \frac{8}{5} = 1.60$$

3. Ans. 5.00



$$\text{for A } T \cos 45^\circ = ma$$

$$\text{for B } mg - T \cos 45^\circ = ma$$

4. Ans. 6.72

Sol. $U = 2x - 4y$

$$F_x = -\frac{\partial U}{\partial x} = -2, F_y = -\frac{\partial U}{\partial y} = 4$$

$$\vec{F} = -2\hat{i} + 4\hat{j}$$

$$\vec{a} = -\hat{i} + 2\hat{j}$$

$$\vec{v} = \vec{u} + \vec{a}t$$

$$\vec{u} = 0 + (-\hat{i} + 2\hat{j}) \times 3$$

$$\vec{v} = -3\hat{i} + 6\hat{j}$$

$$|\vec{u}| = \sqrt{9 + 36} = 3\sqrt{5} = 3 \times 2.24 = 6.72$$

5. Ans. 0.75

Sol. Apply w.e.t. for complete motion

$$w_{\text{external}} + w_g + w_f = 0$$

$$w_{\text{ext}} + 0 - 2\mu mg (AB) = 0$$

$$w_{\text{ext}} = \frac{2\mu mgh}{\tan \theta} = 15$$

$$\text{as } \mu \geq \tan \theta$$

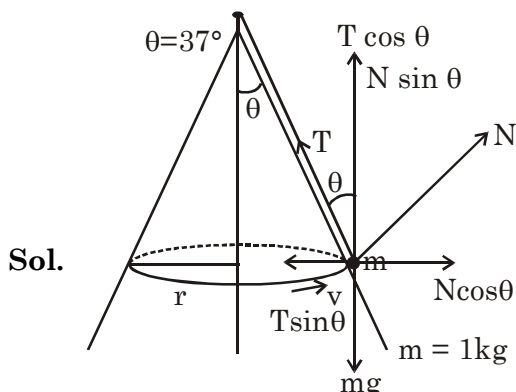
$$2mgh \leq 15$$

$$h \leq \frac{15}{2 \times 1 \times 10}$$

$$h \leq 0.75$$

$$h_{\text{max}} = 0.75$$

6. Ans. 9.80



$$r = \ell \sin \theta$$

$$T \cos \theta + N \sin \theta = mg$$

$$\dots\dots(i)$$

$$T \sin \theta - N \cos \theta = \frac{mv^2}{r}$$

$$T \sin \theta - N \cos \theta = \frac{mv^2}{\ell \sin \theta} \dots\dots(ii)$$

Putting value of θ , ℓ & v

$$\frac{4T}{5} + \frac{3N}{5} = mg \times 4$$

$$\frac{3T}{5} - \frac{4N}{5} = \frac{5}{3} \frac{mv^2}{\ell} \times 3$$

$$5T = 4mg + \frac{5mv^2}{\ell}$$

$$5T = m \left(40 + \frac{5 \times 9}{5} \right)$$

$$5T = m \times 49$$

$$T = \frac{49}{5} = 9.8 \text{ N}$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. **Ans.(D)**

Sol. (A) $B < Mg < Be$ (IE_1) (Correct order)

(B) $NH_3 > PH_3 > AsH_3 > SbH_3$
(Correct order for basic strength)
 $n \uparrow$ basic strength $\downarrow$

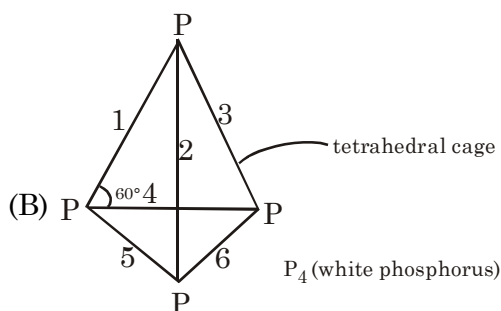
(C) $B < C < N < O$ (E.N.) (Correct order)
in period $Z_{eff} \uparrow$ EN $\uparrow$

(D) IE_1 of Tl $>$ IE_1 of Al (Correct order) due to Lanthanoid contraction

So only statement is correct from given statements

2. **Ans.(B)**

Sol. (A) Black phosphorous is thermodynamically more stable allotrope of phosphorous

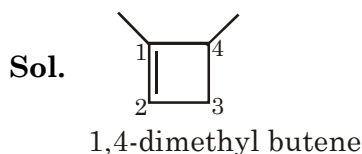


(C) Total 6 p-p bonds

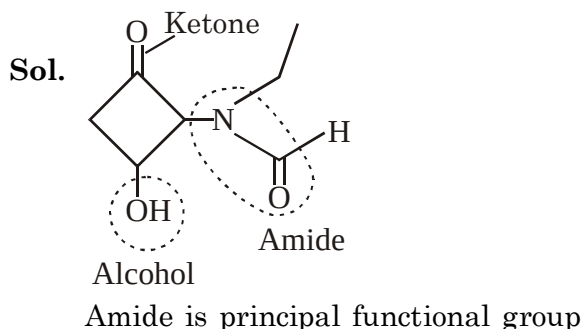
(D) Total 12, 60° bond angles

So only option B is correct from given statements

3. **Ans. (D)**

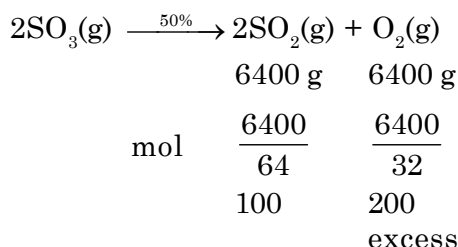


4. **Ans. (D)**



5. **Ans.(A)**

such type of question is solved by excess product



Moles of SO_3 wrt 200 moles of $O_2 = 400$ mol

mass = $400 \times 80 = 3200$

= 32 kg (theoretical)

actual mass of $SO_3 = 32 \times \frac{100}{50} = 64\text{kg}$

6. **Ans.(A)**

A_2 & B_2 are allowed to react in st. coeff. ratio because of maximum energy released

$n_{A_2} = 2/5$

$n_{B_2} = 3/5$

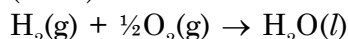
So amount of energy released =

$\frac{2}{5} \times \frac{200}{2} = 40\text{unit}$

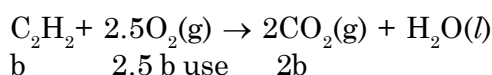
7. **Ans. (A, B, C, D)**

$H_2(g)$ $C_2H_2(g)$ O_2
aml bml 60ml

(a + b) ml



a $\frac{a}{2}$ use



Volume of $O_2 = 13 = 60 - \left(\frac{a}{2} + 2.5b \right)$

$0.5a + 2.5b = 47$

Volume $CO_2 = 45 - 13$

$2b = 32$

$b = 16$ ml

$0.5a + 40 = 47$

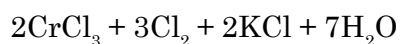
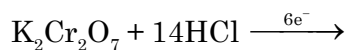
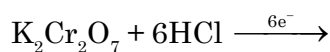
$0.5a = 7$

$a = 14$ ml.

8. Ans.(A,B,C)



$$V_f = 6 \quad V_f = 1$$



9. Ans. (A,B,C,D)

Sol. (A) 2-ethylbutane wrong name due to wrong parent chain selection.

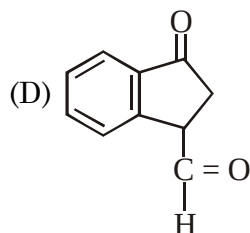
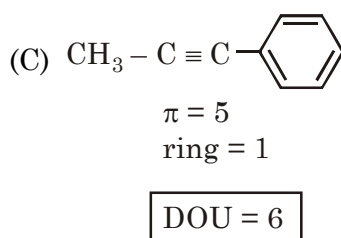
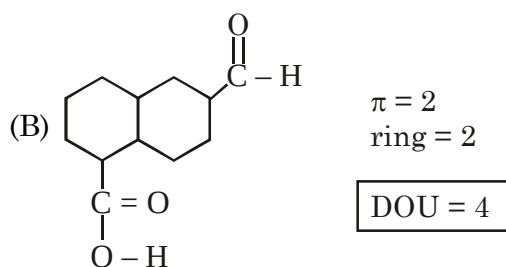
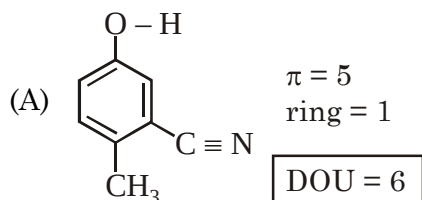
(B) but-1-en-3-ol wrong name due to wrong numbering in parent chain

(C) Wrong name due alphabetic order is not followed in substituents

(D) Wrong name because special suffix cyclopentanoic acid will use

10. Ans. (A,B,C)

Sol.



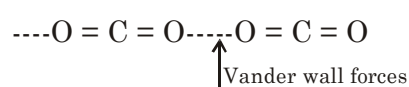
$$\pi = 5$$

$$\text{ring} = 2$$

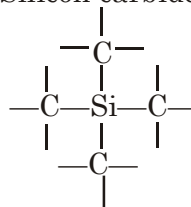
$$\text{DOU} = 7$$

11. Ans.(B,D)

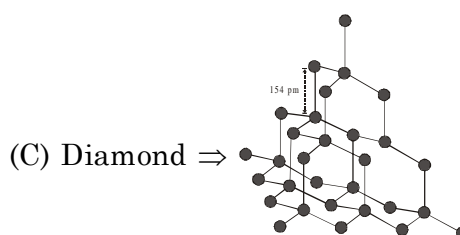
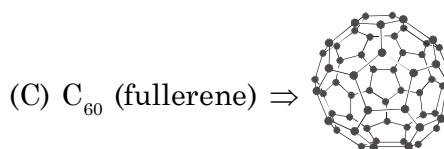
Sol. (A) Dry ice $\rightarrow$ solid CO_2



(B) Silicon carbide (SiC)



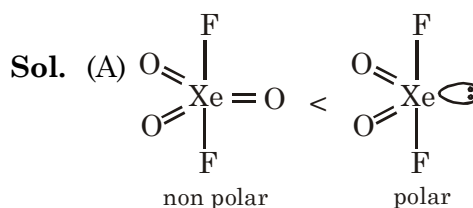
3-D, network structure

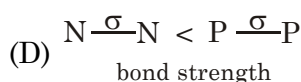
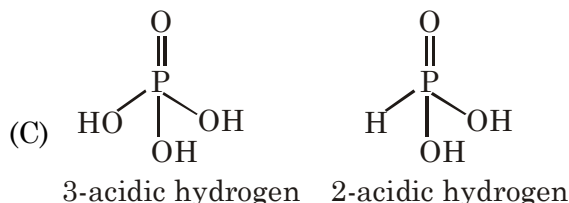
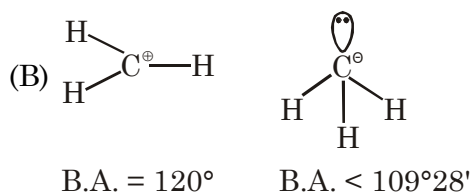


3-D, network structure

So SiC , diamond have 3-D network structure

12. Ans.(A,B,C)





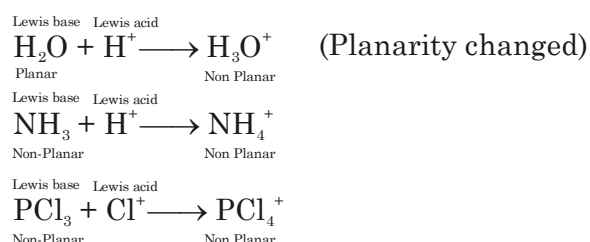
(Due to small size and interelectronic repulsions in N – N)

So A,B,C are correct orders from given options

SECTION-II

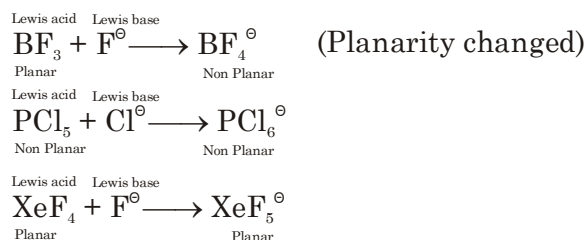
1. Ans.(2.00)

Sol. For calculation of x



So value of x = 1

For calculation of y



So value of y = 1

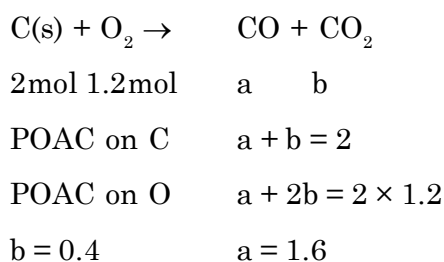
than value of x + y = 1 + 1 = 2

2. Ans. (5.00)

Sol. S, N, Cl, F, C

having more value of electronegativity on Pauling scale as compared to B.

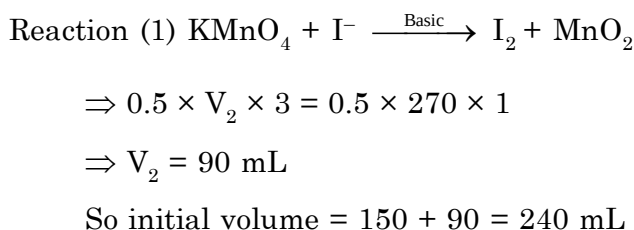
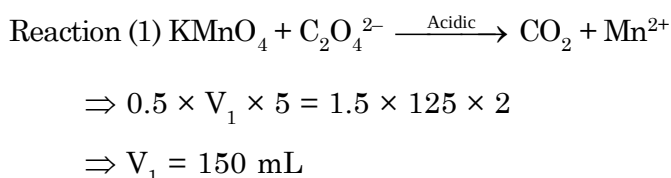
3. Ans.(0.25)



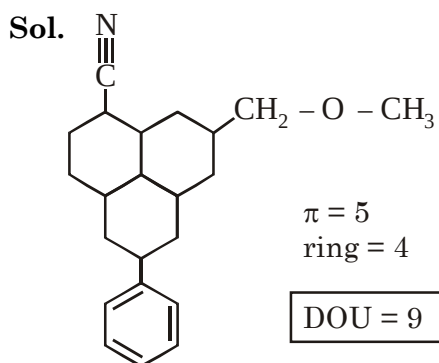
$$\frac{\text{Mole of } CO_2}{\text{Mole of } CO} = \frac{0.4}{1.6} = \frac{1}{4} = 0.25$$

4. Ans. (240.00)

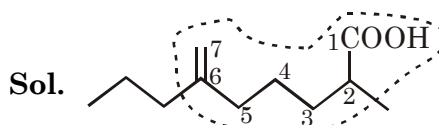
Let V_1 & V_2 are volumes of $KMnO_4$ in 1st part & 2nd part



5. Ans. (9.00)



6. Ans. (7.00)



PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (C)**

$$x = 2 - \sqrt{3} \Rightarrow x^2 - 4x + 1 = 0$$

$$x^3 - 5x^2 + 5x + 1 = x(x^2 - 4x + 1) - (x^2 - 4x + 1) + 2 = 2$$

2. **Ans. (D)**

$\Delta_c = \Delta^{n-1}$; where n is order of determinant

$$\Delta_c = (8)^2 = 64$$

Δ_c is cofactor determinant of Δ

3. **Ans. (B)**

$$\frac{1}{1 - \sin \theta} = 4 + 2\sqrt{3} \Rightarrow 1 - \sin \theta = \frac{4 - 2\sqrt{3}}{4}$$

$$1 - \sin \theta = 1 - \frac{\sqrt{3}}{2}$$

$$\sin \theta = \frac{\sqrt{3}}{2} \Rightarrow \therefore \text{No. of solution in } (-\pi, 3\pi) \text{ are 4.}$$

4. **Ans. (B)**

$$T_r = \frac{r}{(r^4 - 2r^2 + 1) - r^2} = \frac{r}{(r^2 - 1)^2 - r^2} = \frac{r}{(r^2 - 1 + r)(r^2 - 1 - r)}$$

$$T_r = \frac{1}{2} \left[\frac{1}{r^2 - 1 - r} - \frac{1}{r^2 + r - 1} \right]$$

$$T_1 = \frac{1}{2} \left(-\frac{1}{1} - \frac{1}{1} \right)$$

$$T_2 = \frac{1}{2} \left(\frac{1}{1} - \frac{1}{5} \right)$$

$$\vdots \quad \vdots \quad \vdots$$

$$T_n = \frac{1}{2} \left(\frac{1}{n^2 - 1 - n} - \frac{1}{n^2 - 1 + n} \right)$$

$$S_n = \frac{1}{2} \left(-1 - \frac{1}{n^2 - 1 + n} \right)$$

$$\text{as } n \rightarrow \infty, S_\infty = -\frac{1}{2}$$

5. **Ans. (A)**

$$\cos A + \cos C = 2 - 2\cos B$$

$$2\cos\left(\frac{A+C}{2}\right)\cos\left(\frac{A-C}{2}\right) = 4\sin^2 \frac{B}{2}$$

$$2\sin \frac{B}{2} \cos\left(\frac{A-C}{2}\right) = 4\sin^2 \frac{B}{2}$$

$$\underbrace{2\sin \frac{B}{2}}_{\neq 0} \left(\cos\left(\frac{A-C}{2}\right) - 2\sin \frac{B}{2} \right) = 0$$

$$\cos \frac{A}{2} \cos \frac{C}{2} + \sin \frac{A}{2} \sin \frac{C}{2} = 2\sin \frac{B}{2}$$

$$\sqrt{\frac{s(s-a)}{bc}} \sqrt{\frac{s(s-c)}{ab}} + \sqrt{\frac{(s-b)(s-c)}{bc}} \sqrt{\frac{(s-a)(s-c)}{ac}}$$

$$= 2\sqrt{\frac{(s-a)(s-c)}{ac}}$$

$$\frac{s}{b\sqrt{bc}} + \frac{s-b}{b\sqrt{ac}} = \frac{2}{\sqrt{ac}} \Rightarrow a + c = 2b$$

6. **Ans. (D)**

$$D = \begin{vmatrix} \sin A \cos A + \sin A \cos A & \sin A \cos B + \cos A \sin B & \sin A \cos C + \cos A \sin C \\ \sin B \cos A + \cos B \sin A & \sin B \cos B + \sin B \cos B & \sin B \cos C + \cos B \sin C \\ \sin C \cos A + \cos C \sin A & \sin C \cos B + \cos C \sin B & \sin C \cos C + \sin C \cos C \end{vmatrix}$$

$$D = \begin{vmatrix} \sin A & \cos A & 0 \\ \sin B & \cos B & 0 \\ \sin C & \cos C & 0 \end{vmatrix} \begin{vmatrix} \cos A & \cos B & \sin C \\ \sin A & \sin B & \sin C \\ 0 & 0 & 0 \end{vmatrix} = 0$$

7. **Ans. (B,C,D)**

$$\text{Let } x = \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}; x > 0$$

$$x^2 = 2 + x \Rightarrow x = 2$$

$$y = \sqrt{22\sqrt{22\sqrt{22\sqrt{22\dots}}}} \Rightarrow y = \sqrt{22y} \Rightarrow y = 22$$

$$\therefore \log_{\sqrt{2}} \left(\frac{9+2}{22} \right) = \log_{\sqrt{2}} 2^{-1} = -2$$

8. **Ans. (A,B,C,D)**

$$\Delta = \begin{vmatrix} x_1 - y_1 & x_1 - y_2 & x_1 - y_3 \\ x_2 - y_1 & x_2 - y_2 & x_2 - y_3 \\ x_3 - y_1 & x_3 - y_2 & x_3 - y_3 \end{vmatrix}$$

$$R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1$$

$$\Delta = \begin{vmatrix} x_1 - y_1 & x_1 - y_2 & x_1 - y_3 \\ x_2 - x_1 & x_2 - x_1 & x_2 - x_1 \\ x_3 - x_1 & x_3 - x_1 & x_3 - x_1 \end{vmatrix} = 0$$

9. Ans. (B,D)

$$r_1 = \frac{\Delta}{s-a}, r_2 = \frac{\Delta}{s-b}, r_3 = \frac{\Delta}{s-c}$$

$$\frac{s-a}{\Delta}, \frac{s-b}{\Delta}, \frac{s-c}{\Delta} \rightarrow \text{AP}$$

$$a, b, c \rightarrow \text{AP}; 2s = a + b + c = 24$$

$$2b = a + c \Rightarrow 3b = 24 \Rightarrow b = 8$$

$$\therefore a + c = 16 \quad \dots(1)$$

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)} = 24$$

$$24 = \sqrt{12(12-a)(12-8)(12-c)}$$

$$576 = 48(144 - 12(a+c) + ac)$$

$$12 = 144 - 12(16) + ac$$

$$ac = 60 \quad \dots(2)$$

from (1) and (2)

10. Ans. (A,B,D)

$$S_n = 56 \Rightarrow \frac{n}{2}(2a + (n-1)2) = 56$$

$$a - 1 = \frac{56}{n} - n$$

$$\text{since } a \in \mathbb{I} \text{ so } \frac{56}{n} \in \mathbb{I}$$

$$56 = 2^3 \cdot 7^1$$

Hence n can be 2, 4, 8, 7, 14, 28, 56,

11. Ans. (A,B,C)

$$\text{Let } T_1 = 2 = \tan \theta$$

$$T_2 = \frac{\sqrt{3} + T_1}{1 - \sqrt{3}T_1} = \tan\left(\theta + \frac{\pi}{3}\right)$$

$$T_3 = \frac{\sqrt{3} + T_2}{1 - \sqrt{3}T_2} = \tan\left(\theta + \frac{2\pi}{3}\right)$$

$$T_4 = \tan\left(\theta + \frac{3\pi}{3}\right) = \tan(\theta + \pi) = \tan \theta$$

$$\therefore T_1 = T_4 = T_7 = T_{10} = T_{13} = \dots T_{19} = \dots = 2$$

$$T_{20} = T_2 = \tan\left(\theta + \frac{\pi}{3}\right) = \frac{\sqrt{3} + 2}{1 - 2\sqrt{3}} < 0$$

$$T_{18} = T_3 = \tan\left(\theta + \frac{2\pi}{3}\right) = \frac{-\frac{1}{\sqrt{3}} + 2}{1 + \frac{1}{\sqrt{3}} \cdot 2} > 0$$

12. Ans. (B,C)

Divide by $(a^2 + 1)^2$

$$\left(\frac{a^2 + 2}{a^2 + 1}\right)^2 - (b + 4)\left(\frac{a^2 + 2}{a^2 + 1}\right) + b + 3 = 0$$

$$\text{Let } \frac{a^2 + 2}{a^2 + 1} = k$$

$$k^2 - (b + 4)k + b + 3 = 0$$

$$k = 1 \text{ or } k = b + 3$$

$$\text{Now, for } k = 1 \Rightarrow \frac{a^2 + 2}{a^2 + 1} = 1 \Rightarrow \text{No solution}$$

$$\text{for } k = b + 3 \Rightarrow \frac{a^2 + 2}{a^2 + 1} = b + 3$$

$$b + 3 = 1 + \frac{1}{a^2 + 1}$$

$$\text{as } a^2 + 1 \in [1, \infty) \Rightarrow \frac{1}{a^2 + 1} \in (0, 1]$$

$$\therefore 1 < b + 3 \leq 2$$

$$-2 < b \leq -1$$

SECTION-II

1. Ans. 2.00

$$D = \begin{vmatrix} p & p & -1 \\ 1 & p & -1 \\ 1 & p & -p \end{vmatrix} = 0 \Rightarrow p(p-1)^2 = 0$$

$$\therefore p = 0 \text{ or } p = 1.$$

2. Ans. 51.50

$$x^2 + y^2 - 4x - 2y + 5 = 0$$

$$(x-2)^2 + (y-1)^2 = 0 \Rightarrow x = 2 \text{ and } y = 1$$

$$S = \sum_{r=0}^{99} (x + ry) = \sum_{r=0}^{99} (2 + r) = 5150$$

$$\frac{S}{100} = 51.50$$

3. Ans. 4.50

$$\frac{(\sin 1^\circ + \cos 1^\circ)(\sin 2^\circ + \cos 2^\circ) \dots (\sin 89^\circ + \cos 89^\circ)}{(\cos 1^\circ \cos 2^\circ \dots \cos 44^\circ)(\sin 46^\circ \sin 47^\circ \dots \sin 89^\circ) \cos 45^\circ}$$

$$\frac{(\sin 1^\circ + \cos 1^\circ)^2 (\sin 2^\circ + \cos 2^\circ)^2 \dots (\sin 44^\circ + \cos 44^\circ)^2 (\cos 45^\circ + \cos 45^\circ)}{(\cos 1^\circ \cos 2^\circ \cos 3^\circ \dots \cos 44^\circ)^2 \cos 45^\circ}$$

$$\left((1 + \tan 1^\circ)^2 (1 + \tan 2^\circ)^2 \dots (1 + \tan 44^\circ)^2\right) (\sqrt{2})^2$$

$$\left(\underbrace{2^2 \cdot 2^2 \cdot 2^2 \dots 2^2}_{22 \text{ times}}\right) 2 = 2^{45} \Rightarrow k = 45$$

4. Ans. 0.90

$$\sum \alpha = 2 \quad ; \quad \sum \alpha\beta = \frac{1}{2} \quad ; \quad \alpha\beta\gamma = 4$$

$$\frac{1}{\alpha\beta + \gamma - 1} = \frac{1}{\alpha\beta + (2 - \alpha - \beta - 1)} = \frac{1}{\alpha\beta - \alpha - \beta + 1}$$

$$= \frac{1}{(\alpha - 1)(\beta - 1)} = \frac{\gamma - 1}{(\alpha - 1)(\beta - 1)(\gamma - 1)}$$

$$\sum \left(\frac{1}{\alpha\beta + \gamma - 1} \right) = \sum \frac{\gamma - 1}{(\alpha - 1)(\beta - 1)(\gamma - 1)}$$

$$= \frac{\sum \alpha - 3}{\alpha\beta\gamma - \sum \alpha\beta + \sum \alpha - 1} = \frac{-1}{4 - \frac{1}{2} + 1} = \frac{-2}{9}$$

$$\therefore a = \frac{10}{9} \Rightarrow \frac{1}{a} = 0.90$$

5. Ans. 1.50

$$\sin(x + y) = 4 \cos x \cos y$$

$$\sin x \cos y + \cos x \sin y = 4 \cos x \cos y$$

$$\tan x + \tan y = 4 \quad \dots(1)$$

$$\therefore \tan x \tan y = 1 \quad \dots(2)$$

from (1) and (2)

$$\tan x + \frac{1}{\tan x} = 4$$

$$\tan^2 x - 4 \tan x + 1 = 0$$

$$\Rightarrow \tan x = 2 + \sqrt{3} \text{ or } 2 - \sqrt{3}$$

$$\therefore x = \frac{\pi}{12} \text{ or } \frac{5\pi}{12}$$

$$\text{sum} = \frac{\pi}{2}$$

$$\frac{k\pi}{3} = \frac{\pi}{2} \Rightarrow k = \frac{3}{2} = 1.50$$

6. Ans. 1.50

Multiply R_1 by x , R_2 by y and R_3 by z

$$\frac{1}{xyz} \begin{vmatrix} x^4 + x & x^3y & x^3z \\ xy^3 & y^4 + y & y^3z \\ xz^3 & yz^3 & z^4 + z \end{vmatrix} = 11$$

$$\begin{vmatrix} x^3 + 1 & x^3 & x^3 \\ y^3 & y^3 + 1 & y^3 \\ z^3 & z^3 & z^3 + 1 \end{vmatrix} = 11$$

$$R_1 \rightarrow R_1 + R_2 + R_3$$

$$\begin{vmatrix} x^3 + y^3 + z^3 + 1 & x^3 + y^3 + z^3 + 1 & x^3 + y^3 + z^3 + 1 \\ y^3 & y^3 + 1 & y^3 \\ z^3 & z^3 & z^3 + 1 \end{vmatrix} = 11$$

$$x^3 + y^3 + z^3 = 10$$

$$\text{as } x, y, z \in \mathbb{I}^+$$

$$(x, y, z) \equiv (2, 1, 1); (1, 2, 1); (1, 1, 2)$$

$$\therefore k = 3$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
ALLEN COMMON TEST
01-11-2020

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV, V, VI, VII, VIII & IX]

PAPER-2

PART-1 : PHYSICS

SOLUTION

SECTION-I

1. **Ans. (A,B,C)**

Sol. $\mu_1 = \mu_2 \Rightarrow N = 0$ & $a_1 = a_2$

$\mu_1 > \mu_2 \Rightarrow N \neq 0$ & $a_1 = a_2$

$\mu_1 < \mu_2 \Rightarrow N = 0$ & $a_1 \neq a_2$

2. **Ans. (B,C)**

Sol. For maximum value of x , the velocity of block is zero

By W.E.T.

$$\int_0^{x_{\max}} F \cdot dx = k_f - k_i$$

$$\int_0^{x_{\max}} (b - cx) dx = 0 - 0$$

$$\Rightarrow \left(bx - \frac{cx^2}{2} \right)_{x=0}^{x_{\max}=0} \Rightarrow x_{\max} = \frac{2b}{c}$$

Maximum speed at $a = 0$

$$F = 0 \Rightarrow b - cx_0 = 0 \Rightarrow x_0 = \frac{b}{c}$$

$$\int_0^{x_0} F dx = k_f - k_i$$

$$bx_0 - \frac{cx_0^2}{2} = \frac{1}{2}mv_{\max}^2 - 0$$

$$\frac{b^2}{2c} = \frac{1}{2}mv_{\max}^2 \Rightarrow v_{\max} = \frac{b}{\sqrt{mc}}$$

3. **Ans. (A,D)**

Sol. $T + f = m\omega^2 r$

$$0 + \mu mg = m\omega^2 r$$

$$\omega = \sqrt{\frac{\mu g}{r}} = \sqrt{\frac{0.4 \times 10}{1}} = 2 \text{ rad/s}$$

$$T - f = m\omega^2 r$$

For $\omega = 4 \text{ rad/s}$

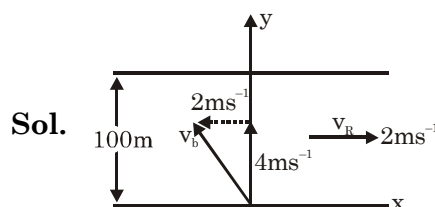
$$T + \mu mg = m\omega^2 r$$

$$T + 0.4 \times 1 \times 10 = 1 \times 1 \times \omega^2$$

$$T + 4 = 16$$

$$T = 12 \text{ N}$$

4. **Ans. (A,B,D)**



Sol.

$$\text{Time of crossing} = \frac{100}{4} = 25 \text{ s}$$

Absolute velocity of boat (i.e. w.r.t. ground)

$$= |\vec{v}_{br} + \vec{v}_r| \quad |\vec{v}_{br} + \vec{v}_r| = |-4\hat{i} + 4\hat{j} + 2\hat{i}|$$

$$= (-2\hat{i} + 4\hat{j}) = \sqrt{20}$$

$$= 2\sqrt{5} \text{ ms}^{-1}$$

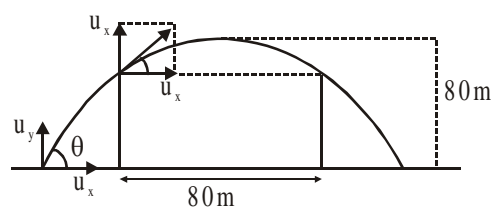
$$\text{Drift of the boat} = -2 \times 25 = -50 \text{ m}$$

5. **Ans. (A,B,C)**

Sol. $mgsin\theta - \mu mgcos\theta \leq F \leq mgsin\theta + \mu mgcos\theta$

6. **Ans. (A,B,C)**

Sol. Time between towers = $\frac{80}{u_x} = \frac{2u_x}{10}$



$$u_x = 20 \therefore t = 4 \text{ s}$$

$$\frac{u_y^2}{2g} = 80 \Rightarrow u_y = 40; T_f = 8 \text{ s}$$

$$R = 160 \text{ m}$$

$$\theta = \tan^{-1} \left(\frac{40}{20} \right)$$

SECTION-II

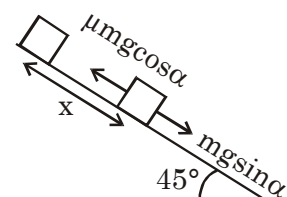
1. Ans. 0.25

Sol. If block stops after distance ℓ using work energy theorem

$$W_{\text{all}} = k_f - k_i$$

$$W_g + W_f = 0 - 0$$

$$(mg \sin \alpha) \ell - \int_0^\ell f_r dx = 0$$



$$mg \ell \sin \alpha - \int_0^\ell 8x mg \cos \alpha dx = 0$$

$$\ell \sin \alpha = \frac{8\ell^2}{2} \cos \alpha \Rightarrow \ell = \frac{1}{4} = 0.25\text{m}$$

2. Ans. 2.12

Sol. KE of part is max when PE is min i.e. at

$$\frac{dU}{dx} = 0 \text{ (stable equilibrium)}$$

$$\frac{dU}{dx} = x^3 - x = 0 \Rightarrow x = 0 \pm 1$$

$$\frac{d^2U}{dx^2} = 3x^2 - 1$$

$\Rightarrow$ at $x = 0$, U is maximum

& at $x = \pm 1$, U is minimum

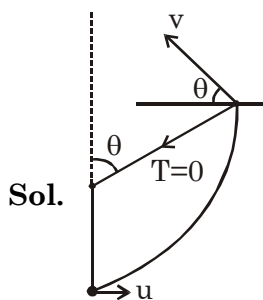
$$\text{at } x = \pm 1, U_{\min} = \frac{1}{4} - \frac{1}{2} = -\frac{1}{4} \text{ J}$$

$$\Rightarrow KE|_{x=\pm 1} = TE - U_{\min}$$

$$= 2 - \left(-\frac{1}{4}\right) = \frac{9}{4} \text{ J} = \frac{1}{2} mv_{\max}^2$$

$$v_{\max} = \sqrt{\frac{9}{2}} = \frac{3}{\sqrt{2}} \text{ m/s}$$

3. Ans. 2.82



Sol.

By energy conservation

$$\frac{1}{2} m \frac{7gr}{2} = \frac{1}{2} mv^2 + mgr(1 + \cos \theta) \quad \dots(i)$$

At the point where string will slack

$$\frac{mv^2}{r} = mg \cos \theta \quad \dots(ii)$$

Solving (i) & (ii) we get

$$\cos \theta = \frac{1}{2} \text{ and } v = \sqrt{\frac{rg}{2}}$$

After string become slack ball performs projectile motion.

Hence $v_{\min} = v \cos \theta$

$$= \frac{\sqrt{rg}}{2\sqrt{2}}$$

$$\text{Hence } p = 2\sqrt{2} = 2.82$$

4. Ans. 12.00

$$\text{Sol. } \frac{dv}{dt} = 6t - 12$$

$$v = 3t^2 - 12t + c$$

$$t = 0, v = 9 \quad \therefore c = 9$$

$$v = 3t^2 - 12t + 9$$

$$x = \int_0^t (3t^2 - 12t + 9) dt$$

$$x = t^3 - 6t^2 + 9t$$

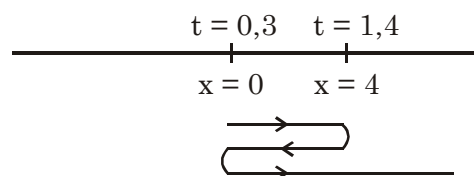
$$v = 3(t^2 - 4t + 3)$$

$$= 3(t^2 - 3t - t + 3)$$

$$= 3[t(t - 3) - 1(t - 3)]$$

$$= 3[(t - 3)(t - 1)]$$

$$v = 0 \text{ at } t = 1, 3$$



5. Ans. 7.66

Sol. $2F - f_s = 0$

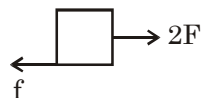
$$2F = \mu_s mg$$

$$2F = 0.5 \times 24 \times 10$$

$$2F = 120$$

$$F = 60 \text{ N} = kt_0^2$$

$$t_0 = \frac{60}{k} = 1 \text{ sec}$$



$$1 \text{ s} \leq t \leq 2 \text{ s}$$

$$2F - f_k = ma$$

$$\frac{dv}{dt} = a = \frac{2kt^2 - 0.4 \times 240}{24}$$

$$\frac{dv}{dt} = \frac{2 \times 60t^2 - 96}{24} = 5t^2 - 4$$

$$\int_0^v dv = \left[\frac{5t^3}{3} - 4t \right]_1^2$$

$$v = \frac{5(8)}{3} - 8 - \left(\frac{5}{3} - 4 \right)$$

$$v = \frac{40}{3} - 8 - \frac{5}{3} + 4$$

$$v = \frac{35}{3} - 4$$

$$v = \frac{35 - 12}{3} = \frac{23}{3}$$

$$3 = \frac{23}{v} \Rightarrow v = 7.66$$

6. Ans. 0.02

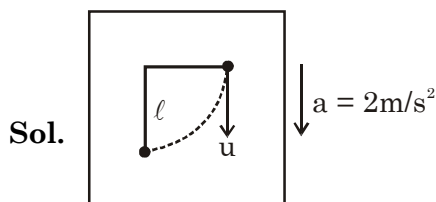
Sol. $4.9 \text{ LT}^{-2} = 9.8 \text{ ms}^{-2}$

$$100 \text{ ML}^2\text{T}^{-2} = \frac{1}{2} \times 20(4)^2 \text{ kgm}^2\text{s}^{-2}$$

$$10 \text{ MLT}^{-1} = 20 \times 4 \text{ kg ms}^{-1}$$

SECTION-III

1. Ans. 3



Sol.

$$l = 2 \text{ m}$$

$$g_e = g - a$$

$$= 10 - 2 = 8 \text{ m/s}^2$$

$$u = \sqrt{3g_e l} = \sqrt{3 \times 8 \times 2}$$

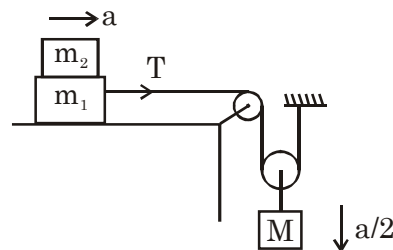
$$= \sqrt{48} = 4\sqrt{3}$$

2. Ans. 4

Sol. $Mg - 2T = M\left(\frac{a}{2}\right) \quad \dots (i)$

$$T = (m_1 + m_2)a \quad \dots (ii)$$

From equation (i) and (ii)



$$Mg = \left(\frac{M}{2} + 2(m_1 + m_2) \right) a$$

$$a = \frac{2Mg}{M + 4(m_1 + m_2)}$$

m_2 will slide when

$$\mu m_2 g = m_2 a$$

$$5 = \frac{2Mg}{M + 12}$$

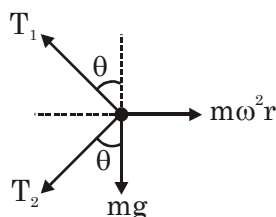
$$5M + 60 = 20M$$

$$15M = 60$$

$$M = 4$$

3. Ans. 5

Sol. FBD of m is rotating frame



For equilibrium of bob in rotating frame

$$T_1 \sin \theta + T_2 \sin \theta = m\omega^2 r \quad \dots (i)$$

$$T_1 \cos \theta = T_2 \cos \theta + mg \quad \dots (ii)$$

from (ii)

$$T_1 - T_2 = \frac{mg}{\cos \theta} = \frac{2mg\ell}{h} \left[\cos \theta = \frac{h}{2\ell} \right] \quad \dots (iii)$$

from (i)

$$T_1 + T_2 = \frac{m\omega^2 r}{\sin \theta} = m\omega^2 \ell \left[\sin \theta = \frac{r}{\ell} \right]$$

$$T_1 + T_2 = m\ell \left(\frac{8g}{h} \right) = \frac{8mg\ell}{h} \quad \dots (iv)$$

$$(iii)/(iv) \Rightarrow \frac{T_1 - T_2}{T_1 + T_2} = \frac{1}{4}$$

$$4T_1 - 4T_2 = T_1 + T_2$$

$$3T_1 = 5T_2$$

$$\frac{T_1}{T_2} = \frac{5}{3} \text{ ans.}$$

4. Ans. 3

Sol. If finally elongation in spring is y, to break off lower block $ky = mg$

If initial compression is x, by W/E th

$$w_g + w_{sp} = \Delta k$$

$$-mg(x + y) + \frac{1}{2} kx^2 - \frac{1}{2} ky^2 = 0$$

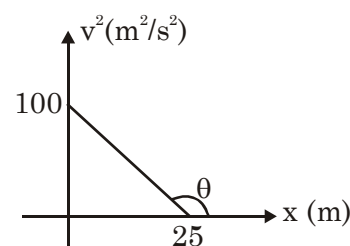
$$-kyx - ky^2 + \frac{1}{2} kx^2 - \frac{1}{2} ky^2 = 0$$

$$x^2 - 2xy - 3y^2 = 0$$

$$x = \frac{2y + \sqrt{4y^2 + 12y^2}}{2}$$

$$x = 3y = \frac{3mg}{k}$$

5. Ans. 2



Sol.

$$m = \tan \theta = -4$$

Equation of straight line

$$y = mx + C$$

$$v^2 = -4x + 100$$

differentiate

$$2v \frac{dv}{dx} = -4$$

$$a = v \frac{dv}{dx} = -2m/s^2$$

$$|a| = 2 \text{ m/s}^2$$

6. Ans. 1

$$\text{Sol. } 10 + 16t + \frac{1}{2} \times 1 \times t^2 = 12t + \frac{1}{2} \times 2 \times t^2$$

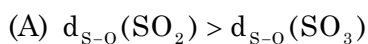
$$t = 10 \text{ sec.}$$

PART-2 : CHEMISTRY

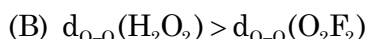
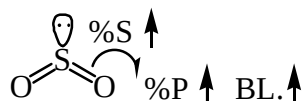
SOLUTION

SECTION-I

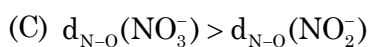
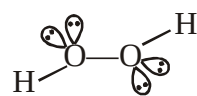
1. Ans.(A,B,C)



According to Bent's rule



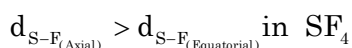
According to Bent's rule



$$B.O. = \frac{4}{3} < B.O. = \frac{3}{2}$$

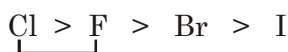
B.O. $\uparrow$ BL $\downarrow$

(D) As per bents rule

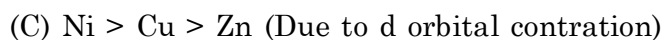
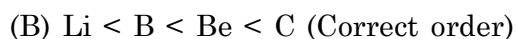


2. Ans.(A, D)

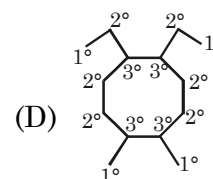
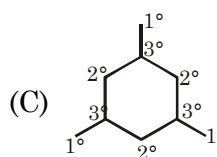
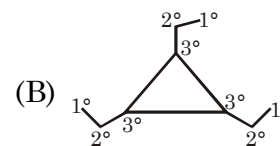
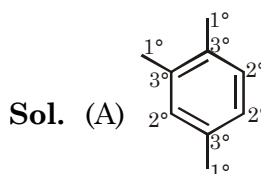
Sol.



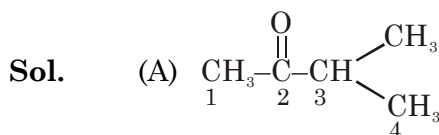
Due to interelectronic repulsion



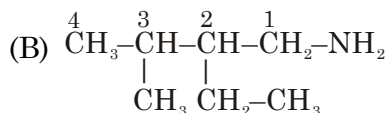
3. Ans. (A,B,C)



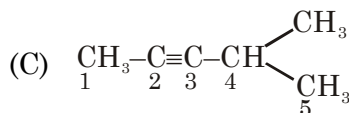
4. Ans. (A,B,D)



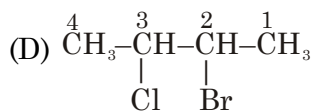
3-Methyl-2-Butanone



2-Ethyl-3-Methyl Butanamine



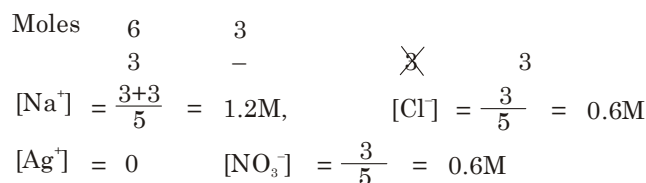
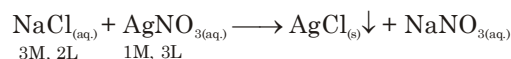
4-Methyl-2-Pentyne



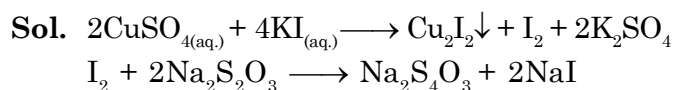
2-Bromo-3-Chlorobutane

5. Ans.(A,C)

Sol.



6. Ans.(A,C,D)



$$\text{mole of I}_2 = \frac{\text{moles of Na}_2\text{S}_2\text{O}_3}{2} = \frac{0.1 \times 0.2}{2} = 0.01$$

$$\eta_{\text{CuSO}_4} = 2\eta_{\text{I}_2} = 0.02$$

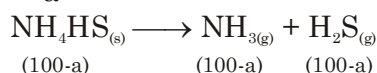
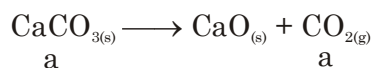
$$\text{wt. CuSO}_4 = 0.02 \times 160 = 3.2\text{g}$$

SECTION-II

1. Ans.(50.00)

Sol. Total 100 mol (assume)

$$\text{mol CaCO}_3 = a \quad \eta_{\text{NH}_4\text{HS}} = 100 - a$$



Molecular weight of gaseous mixture

$$= \frac{95}{3} = \frac{44a + 17(100 - a) + 34(100 - a)}{200 - a}$$

$$a = 50$$

2. Ans.7.50

$$\left(\frac{10}{11.35} + \frac{1.5}{11.35} + \frac{20}{11.35} \right) \times 500 = M_f \times 3000$$

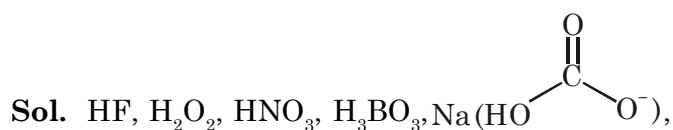
$$M_f = \frac{45}{11.35 \times 6}$$

$$\text{Vol. strength of final solution} = M \times 11.35$$

$$= \frac{45}{11.35 \times 6} \times 11.35$$

$$= 7.5 \text{ V}$$

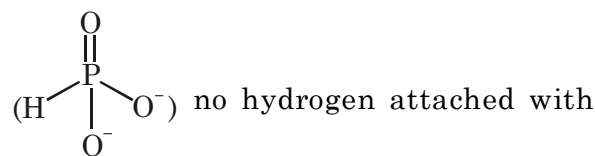
3. Ans.(6.00)



$\text{HF}_2^\ominus$ ($\text{F}^\ominus \cdots \text{H} - \text{F}$) shows hydrogen bonding.

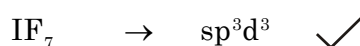
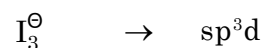
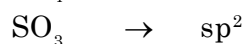
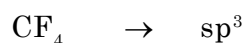
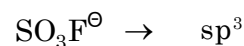
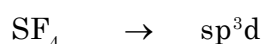
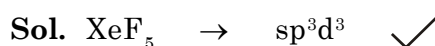
So total 6 species shows H bonding.

In Chloral ($\text{Cl}_3\text{C}.\text{CHO}$) and HPO_3^{-2}

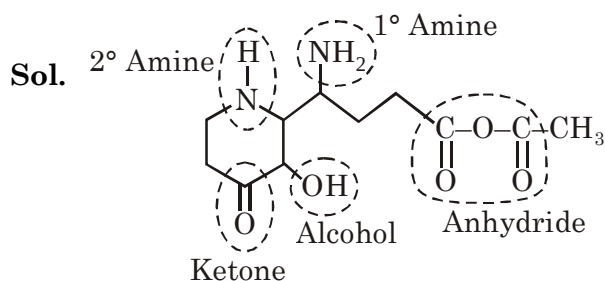


(F,N,O) so no hydrogen bonding.

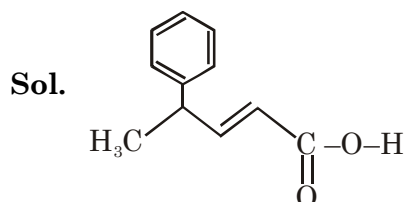
4. Ans.(3.00)



5. Ans. (5.00)



6. Ans. (6.00)



$$\pi = 5$$

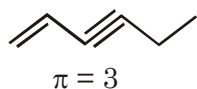
$$\text{ring} = 1$$

$$\text{DBE} = 6$$

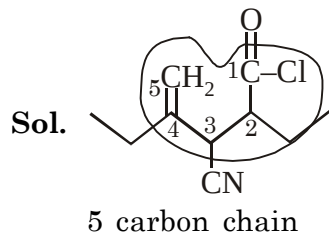
SECTION-III

1. Ans. (3)

Sol. 1-hexen-3-yne

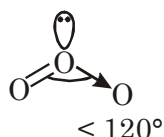


2. Ans. (5)

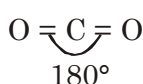


3. Ans. (2)

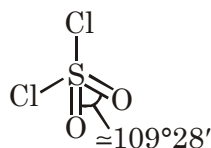
Sol. O_3



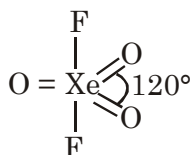
CO_2



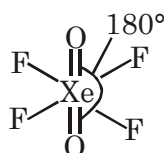
SO_2Cl_2



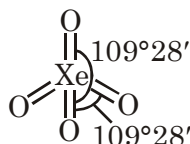
XeO_3F_2



XeO_2F_4

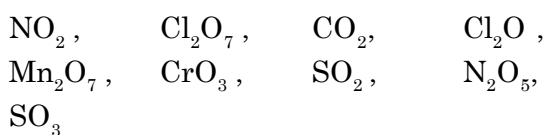


XeO_4



4. Ans. (9)

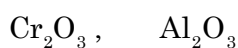
Sol. Acidic oxides



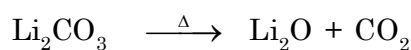
Neutral oxide



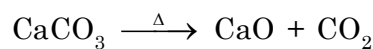
Amphoteric oxides



5. Ans. (2)



$$(248 - x) \text{ gm} = \frac{(248 - x)}{74} \text{ mole}$$



x gm

$$\Rightarrow \frac{x}{100} \text{ mole}$$

$$\text{So } n_{CO_2} \text{ produced} = \frac{x}{100} + \frac{(248 - x)}{74}$$

$$\text{Now } PV = nRT$$

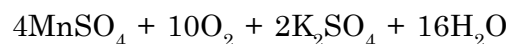
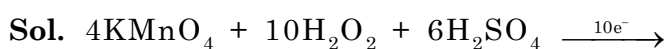
$$3 \times 24 = \left(\frac{x}{100} + \frac{248 - x}{74} \right) \times 0.08 \times 300$$

$$\Rightarrow x = 100 \text{ g} \quad \Rightarrow n_{CaCO_3} = 1$$

$$n_{Li_2CO_3} = \frac{(248 - 100)}{74} = 2$$

$$\text{So, } \frac{n_{Li_2CO_3}}{n_{CaCO_3}} = 2$$

6. Ans. (6)



$$x = 4, z = 6, p = 4$$

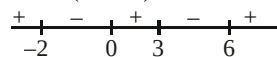
PART-3 : MATHEMATICS**SOLUTION****SECTION-I**1. **Ans. (A,B)**

$$\text{We know } 2R_{ABC} = \frac{a}{\sin A}$$

$$\text{and } R_{ABC} = 2R_{DEF}$$

2. **Ans. (A,C)**

$$\frac{(x+2)(x-3)}{x(x-6)} \leq 0$$



$$x \in [-2, 0) \cup [3, 6)$$

then integral values of x are $-2, -1, 3, 4, 5$

3. **Ans. (A,C)**

$$N = \underset{\text{integral part}}{6^{10}} + \underset{\text{fractional part}}{6^{-10}}$$

so number of digits can be found by taking log at the base 10

$$\log_{10} 6^{10} = 10[\log_{10} 2 + \log_{10} 3]$$

$$= 10[.3010 + .4771]$$

$$= 10[.7781] = 7.781$$

$$\text{characteristic} = 7$$

$$\Rightarrow \text{No. of digits before decimal} = 8$$

$$\text{similarly characteristic of } 6^{-10} = -8$$

$$\text{so number of zeroes after decimal} = 7$$

4. **Ans. (B,D)****Method-1**

Here

$$\text{Let } L_1 \Rightarrow kx + y + z = 2$$

$$L_2 \Rightarrow (k+1)x + 2y + 2z = 4$$

$$\text{and } L_3 \Rightarrow x + y + z = 2$$

$$\text{here } L_2 = L_1 + L_3$$

so we can say actually we have only 2 different lines or equations but 3 variables at $k = 4, 1, -1$, so ∞ solutions.

Method-2

Here

$$D = D_1 = D_2 = D_3 = 0 \text{ for } k = 1, -1, 4$$

so infinitely many solutions.

Method-3

$$\text{as } L_2 = L_1 + L_3$$

$$\Rightarrow \text{family of concurrent lines}$$

so always have a solution for any k .

5. **Ans. (A,C)**

$$1 + \frac{1}{2} + \frac{1}{2^2} + \dots = \frac{1}{1 - \frac{1}{2}} = 2$$

similarly

$$\text{let } S = 1.\bar{9}$$

$$\Rightarrow S = 1.999\dots\infty \quad \dots(1)$$

$$\Rightarrow 10S = 19.99\dots\infty \quad \dots(2)$$

Now equation (2) .. equation (1)

$$\Rightarrow S = 2$$

6. **Ans. (C)**

Here

$$\left(\frac{a_1x}{1} - \frac{a_2}{2}\right)^2 + \left(\frac{a_2x}{2} - \frac{a_3}{3}\right)^2 + \dots = 0$$

$$\Rightarrow x = \frac{a_2/2}{a_1/1} = \frac{a_3/3}{a_2/2} = \dots$$

$$\text{so } \frac{a_1}{1}, \frac{a_2}{2}, \frac{a_3}{3}, \dots \text{ are in G.P.}$$

SECTION-II1. **Ans. 0.33**

$$y = \frac{x^2 + x + 1}{x^2 - x + 1} \quad x \in \mathbb{R}$$

$$\Rightarrow (y-1)x^2 - (y+1)x + y-1 = 0$$

$$D \geq 0$$

$$\Rightarrow (y+1)^2 - 4(y-1)^2 \geq 0$$

$$\Rightarrow (y-3)\left(y - \frac{1}{3}\right) \leq 0$$

$$\Rightarrow y \in \left[\frac{1}{3}, 3\right]$$

$$\text{so minimum value of } y = \frac{1}{3} = 0.33\dots$$

2. **Ans. 8.00**

$$\text{if } a, b, c \geq 0$$

$$\text{then } AM \geq GM$$

$$\Rightarrow \frac{a+b+c}{3} \geq (abc)^{1/3}$$

$$\Rightarrow \frac{6}{3} \geq (abc)^{1/3}$$

$$\Rightarrow abc \leq 8$$

3. **Ans. 2.18 or 2.19**

$$\begin{aligned} & 2 \left[\sin^8 \frac{\pi}{16} + \sin^8 \frac{3\pi}{16} + \sin^8 \frac{5\pi}{16} + \sin^8 \frac{7\pi}{16} \right] \\ &= 2 \left[\sin^8 \frac{\pi}{16} + \cos^8 \frac{\pi}{16} + \sin^8 \frac{3\pi}{16} + \cos^8 \frac{3\pi}{16} \right] \\ &= 2 \left[\left(\cos^4 \frac{\pi}{16} - \sin^4 \frac{\pi}{16} \right)^2 + 2 \sin^4 \frac{\pi}{16} \cos^4 \frac{\pi}{16} \right. \\ &\quad \left. + \left(\cos^4 \frac{3\pi}{16} - \sin^4 \frac{3\pi}{16} \right)^2 + 2 \sin^4 \frac{3\pi}{16} \cos^4 \frac{3\pi}{16} \right] \\ &= 2 \left[\cos^2 \frac{\pi}{8} + \frac{1}{8} \sin^4 \frac{\pi}{8} + \cos^2 \frac{3\pi}{8} + \frac{1}{8} \sin^4 \frac{3\pi}{8} \right] \\ &= 2 \left[\cos^2 \frac{\pi}{8} + \sin^2 \frac{\pi}{8} + \frac{1}{8} \left[\sin^4 \frac{\pi}{8} + \cos^4 \frac{\pi}{8} \right] \right] \\ &= 2 \left[1 + \frac{1}{8} \left(1 - 2 \sin^2 \frac{\pi}{8} \cos^2 \frac{\pi}{8} \right) \right] \\ &= 2 \left[1 + \frac{1}{8} \left(1 - \frac{1}{2} \sin^2 \frac{\pi}{4} \right) \right] \\ &= 2 \left[1 + \frac{1}{8} \left(\frac{3}{4} \right) \right] \\ &= 2 \left[1 + \frac{3}{32} \right] = \frac{35}{16} = 2.18 \text{ or } 2.19 \end{aligned}$$

4. **Ans. 2.00**

$$f(2) < 0 \text{ and } f(4) < 0$$

5. **Ans. 19.90**

Let country (C_1) has x population and average age 100 years

$$\Rightarrow \text{so total age of all persons of } C_1 = 100x$$

$$\text{similarly for } C_2 \Rightarrow 90(2x)$$

$$\text{similarly for } C_3 \Rightarrow 80(2^2x)$$

so average age of all the persons

$$= \frac{100x + 90(2x) + 80(2^2x) + 70(2^3x) + \dots + 10(2^9x)}{x + 2x + 2^2x + \dots + 2^9x}$$

$$= 19.90$$

[Here numerator is arithmetics geometric series and denominator is a geometric series]

6. **Ans. 2.00**

$$\begin{aligned} & \sum_{r=1}^{2n} (r(r+1))^2 (-1)^r \\ &= -\underbrace{(1.2)^2 + (2.3)^2}_{\dots} - \underbrace{(3.4)^2 + (4.5)^2}_{\dots} - \dots - 2n \text{ terms} \end{aligned}$$

(here a means product)

$$= 2^2[2.4] + 4^2[2.8] + 6^2[2.12] \dots n \text{ terms}$$

$$= 2^3[1^2.4 + 2^2.8 + 3^2.12 \dots n \text{ terms}]$$

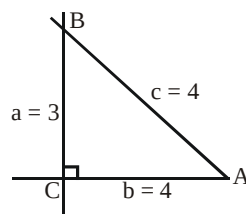
$$= 2^5[1^3 + 2^3 + 3^2 + \dots n \text{ terms}]$$

$$= 2^5 \left(\frac{n(n+1)}{2} \right)^2$$

similarly solve numerator

SECTION-III

1. **Ans. 1**



$$\text{here } \Delta = \frac{1}{2} \times 3 \times 4 = 6$$

$$s = \frac{3+4+5}{2} = 6$$

$$\text{so } r = \frac{\Delta}{s} = \frac{6}{6} = 1$$

2. **Ans. 3**

length of internal angle bisector of $\angle A$

$$= \frac{2bc}{b+c} \cos \frac{A}{2} = \frac{2 \times 4 \times 12}{4+12} \cos \frac{\pi}{3}$$

$$= 3$$

3. **Ans. 1**

$$(\sin x - 2) \left(\cos x - \frac{1}{2} \right) \left(\cos x - \frac{\sqrt{3}}{2} \right) \geq 0$$

$$\text{but } (\sin x - 2) < 0$$

$$\Rightarrow \left(\cos x - \frac{1}{2} \right) \left(\cos x - \frac{\sqrt{3}}{2} \right) \leq 0$$

$$\Rightarrow \cos x \in \left[\frac{1}{2}, \frac{\sqrt{3}}{2} \right]$$

$$\Rightarrow x \in \left[\frac{\pi}{6}, \frac{\pi}{3} \right] \cup \left[\frac{5\pi}{3}, \frac{11\pi}{6} \right] \text{ but } \pi \approx 3.14$$

$$\approx (.52 \text{ to } 1.05) \text{ or } (5.23 \text{ to } 5.76)$$

so there is only one integral value of $x = 1$

4. **Ans. 2**

$$3(\sin^3 x - \cos^3 x) - \cos 2x (2 + \sin 2x) = 0$$

$$\Rightarrow 3(\sin x - \cos x) (1 + \sin x \cos x) + (\sin^2 x - \cos^2 x) (2 + \sin 2x) = 0$$

$$\Rightarrow (2 + \sin 2x) \left[\frac{3}{2} (\sin x - \cos x) + (\sin^2 x - \cos^2 x) \right] = 0$$

$$\Rightarrow (\sin x - \cos x) \left[\frac{3}{2} + \sin x + \cos x \right] [2 + \sin 2x] = 0$$

$$\text{but } 2 + \sin 2x \neq 0 \text{ and } \sin x + \cos x \neq -\frac{3}{2}$$

$$(\text{because } -\sqrt{2} \leq \sin x + \cos x \leq \sqrt{2})$$

$$\Rightarrow \sin x = \cos x \Rightarrow \tan x = 1 \Rightarrow \text{number of solutions in } [0, 2\pi] \text{ is } 2$$

5. **Ans. 4**

$$\Rightarrow -\cos 2x - 2k \sin x + 4k - 7 = 0$$

$$\Rightarrow -1 + 2\sin^2 x - 2k \sin x + 4k - 7 = 0$$

$$\Rightarrow 2\sin^2 x - 2k \sin x + 4k - 8 = 0$$

$$\Rightarrow \sin^2 x - k \sin x + 2k - 4 = 0$$

$$\sin^2 x - 4 - k(\sin x - 2) = 0$$

$$(\sin x - 2) [\sin x + 2 - k] = 0$$

$$\sin x \neq 2$$

$$\Rightarrow \sin x = k - 2$$

$$\Rightarrow -1 \leq k - 2 \leq 1$$

$$\Rightarrow 1 \leq k \leq 3$$

$$\Rightarrow k \in [1, 3]$$

6. **Ans. 3**

$$\frac{1}{7} \left[\left(3x^2 + \frac{4}{3x^2} + 1 \right)^2 - 2 \left(3x^2 + \frac{4}{3x^2} + 1 \right) + 6 \right]$$

$$\text{Let } t = 3x^2 + \frac{4}{3x^2} + 1 \geq 5$$

$$\text{Now } \frac{1}{7} [t^2 - 2t + 6] = \frac{1}{7} [(t-1)^2 + 5]$$

$$\text{Now } t \in [5, \infty], \quad (t-1)^2 \in [16, \infty]$$

$$\Rightarrow t-1 \in [4, \infty], \quad (t-1)^2 + 5 \in [21, \infty)$$

$$\frac{1}{7} [(t-1)^2 + 5] \in [3, \infty)$$